Investigating the Effect of Path-Length Difference on the Spectral Response of Mach-Zehnder Interferometers and Extracting the Group Index

Oct 25, 2026 • Mukul Sharma

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Submitted Layout file name: EBeam_0nmPhoton.gds

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Introduction

Silicon photonics uses silicon waveguides, only a few hundred nanometres across, to guide light on a chip made with the same fabrication methods as electronic circuits. Many devices, such as filters, delay lines and sensors, depend on how fast light travels in the waveguide, which is set by its group index. The group index is very sensitive to the waveguide's width and thickness, so it differs slightly from chip to chip. This makes it important to design devices with a good estimate of the group index, and then to measure it on the fabricated chip.

A Mach-Zehnder interferometer (MZI) is a simple device for this. Light is split into two waveguide arms, travels different distances, and is then recombined. The two beams interfere, which gives a transmission spectrum of regular peaks and nulls. If the arms differ in length by ΔL, the spacing between neighbouring peaks, the free spectral range (FSR), is $FSR = \lambda^2/(n_g \cdot \Delta L)$, where λ is the wavelength and n_g is the group index [1]. A longer ΔL gives more closely spaced fringes. So the FSR measured from a fabricated MZI gives the group index of the waveguide directly.

This report presents the design and simulation of four unbalanced MZIs with ΔL = 50.000, 100.000, 300.000 and 500.000 µm, together with a loopback reference, for a 500 nm × 220 nm silicon strip waveguide at 1550 nm. The group index of the TE fundamental mode is calculated with a mode solver, the expected FSR of each MZI is predicted from it, and the simulated spectra of the circuits are compared with that prediction. After fabrication, the group index will be extracted from the measured spectra and compared with the simulated value.

Theory

The fringe spacing of an unbalanced MZI is $FSR = \lambda^2/(n_g \cdot \Delta L)$, so a measured FSR and a known ΔL give the group index. This section shows where that relation comes from.

Light propagating through a waveguide arm of length L accumulates a phase $\varphi = \beta L$, where $\beta = 2\pi n_{eff}/\lambda$ and n_{eff} is the effective index of the mode. For an MZI with arm lengths L_1 and L_2, the phase difference between the two arms is

$$\Delta\varphi = \beta_2 L_2 - \beta_1 L_1$$

$$\Delta\varphi = \frac{2\pi}{\lambda}\left(n_{eff,2}.L_2 - n_{eff,1}.L_1\right) \quad (1)$$

Both arms use the same waveguide, so $n_{eff,1} = n_{eff,2} = n_{eff}$. With $\Delta L = L_2 - L_1$ this becomes

$$\Delta\varphi = 2\pi.n_{eff}.\frac{\Delta L}{\lambda} \quad (2)$$

At the output coupler the two paths interfere. One output has a maximum where $\Delta\varphi = 2m\pi$ and a null where $\Delta\varphi = (2m+1)\pi$, with m an integer, and the other output behaves the opposite way. This shows up as peaks and nulls in the spectrum, and the distance in wavelength between neighboring peaks (or neighboring nulls) of one output is the free spectral range (FSR).

Moving from one fringe to the next changes $\Delta\varphi$ by 2π. Differentiating Eq. (2) with respect to wavelength gives

$$\frac{d\Delta\varphi}{d\lambda} = -\left(\frac{2\pi\Delta L}{\lambda^2}\right)\left(n_{eff} - \frac{\lambda\,.\,dn_{eff}}{d\lambda}\right) \quad (3)$$

Setting $|d\Delta\phi/d\lambda|.\ FSR = 2\pi$ gives

$$FSR = \frac{\lambda^2}{n_g\,.\Delta L} \quad (4)$$

where the group index is defined as

$$n_g = \left(n_{eff} - \frac{\lambda\cdot dn_{eff}}{d\lambda}\right) \quad (5)$$

Extracting n_g from spectra

1. Record the output spectrum of each MZI over the full wavelength window around 1550 nm.
2. Locate each null by fitting a parabola to the points around its minimum. The spacing between adjacent nulls of the same output is the local FSR, assigned to their midpoint wavelength. Because n_g depends on wavelength, the FSR is evaluated locally.

3. Fit the local FSR against wavelength with a quadratic polynomial and evaluate the fit at 1550 nm. Then calculate $n_g = \lambda^2/(FSR \cdot \Delta L)$ for each device.
4. Plot 1/FSR against ΔL for the four devices. Since $1/FSR = n_g \cdot \Delta L/\lambda^2$, the slope is n_g/λ^2, and a free intercept absorbs any constant extra path length that is the same in every device.

Modelling and simulation

With the group index of 4.215 from the mode simulation, the four designs have expected FSRs of 11.40, 5.70, 1.90 and 1.14 nm (Table 1), so every device shows at least eight fringes within the 100 nm measurement bandwidth.

The waveguide is a 500 nm × 220 nm silicon strip with oxide cladding, TE-polarised, with a bend radius of 5 µm.

Table 1. Arm lengths of the four unbalanced MZIs. L1 and L2 are the lengths of the short and long arms, and ΔL = L2 − L1. The expected FSR at 1550 nm is calculated from $FSR = \lambda^2/(n_g \cdot \Delta L)$ with $n_g = 4.215$, and the last column is the number of fringes expected in a 100 nm window.

Design	L1 (µm)	L2 (µm)	ΔL (µm)	Expected FSR (nm)	Fringes in 100 nm
MZI1	47.678	97.678	50.000	11.40	9
MZI2	47.678	147.678	100.000	5.70	18
MZI3	47.678	347.678	300.000	1.90	53
MZI4	47.678	547.678	500.000	1.14	88

The arm lengths are those of the waveguides in the KLayout layout, as extracted into the INTERCONNECT circuit. The expected FSR uses $n_g = 4.215$ from the mode solution. At the 20 pm wavelength step used in the simulation, even MZI4, which has the narrowest fringes, has about 57 points per fringe (about 114 at a 10 pm laser step).

The waveguide was modelled in Lumerical MODE as a 500 nm wide, 220 nm thick silicon strip surrounded by silicon dioxide (Figure 1). The mode solver was run at 1550 nm on a 10 nm mesh; halving the mesh to 5 nm changed the group index by less than 0.03%, so the 10 nm mesh was kept. Silicon and silicon dioxide were described by the Palik data fitted with a multi-coefficient model over 1.5 to 1.6 µm (RMS error 0.0014 and 1.9×10^{-5} respectively), because the raw tabulated silicon data is too sparse near 1.5 µm and produces an artificial step in n_g.

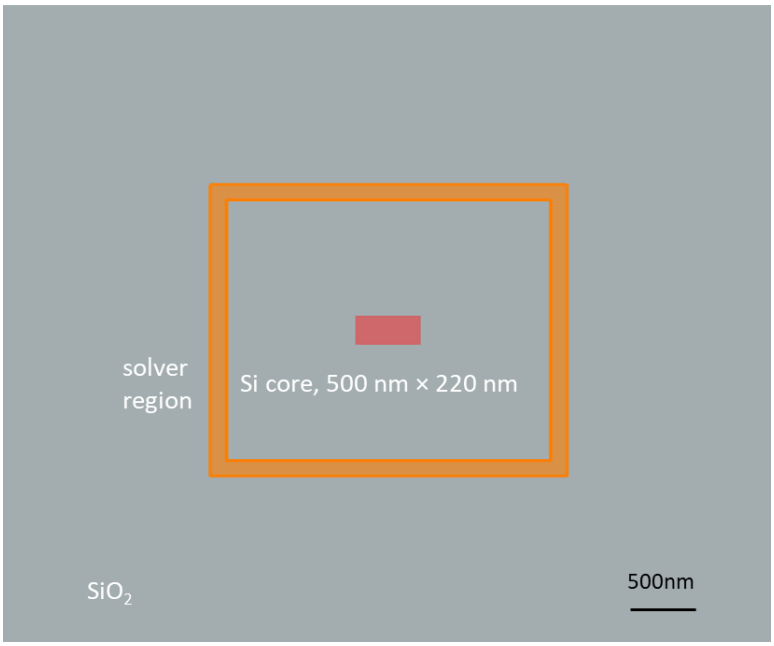


Figure 1. Cross-section (Y–Z plane) of the waveguide model in MODE: a 500 nm × 220 nm silicon core (pink) in silicon dioxide (grey). The orange frame marks the 2.5 µm × 2.0 µm mode-solver region.

The mode solver finds three modes at 1550 nm (Figure 2). Mode 1 is the fundamental quasi-TE mode. Its electric field is mostly polarised along the width of the waveguide and is tightly confined in the silicon core, with an effective index of 2.445 and a TE fraction of 98%. Mode 2 is TM-like, with a TE fraction of only 4% and an effective index of 1.77. Mode 3 (n_{eff} = 1.49, only slightly above the oxide index) is a weakly guided higher-order mode near cutoff, with much of its field in the cladding; it is not excited by the TE grating couplers. Therefore, only Mode 1 is used in the rest of this work, since the grating couplers excite TE light, and it was tracked through the wavelength sweep.

Figure 3 shows the effective index (a) and group index (b) of the fundamental TE mode from 1500 to 1600 nm. The effective index decreases smoothly with wavelength, from about 2.50 at 1500 nm to 2.39 at 1600 nm. The group index is larger than the effective index and increases from 4.163 to 4.275 over the same range, with a value of 4.216 at 1550 nm. The effective index as a function of wavelength was fitted over this range with a quadratic polynomial:

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$$

with λ_0 = 1.55 µm, n_1 = 2.44532, n_2 = −1.1421 µm^{-1} and n_3 = −0.3613 µm^{-2}. The group index follows from Eq. (5), $n_g = n_{eff} - \lambda \cdot dn_{eff}/d\lambda$ (which comes from $n_g = c \cdot d\beta/d\omega$ with $\beta = 2\pi n_{eff}/\lambda$). For the fit this gives

$$n_g(\lambda) = n_1 - \lambda_0 n_2 - 2\lambda_0 n_3(\lambda - \lambda_0) - n_3(\lambda - \lambda_0)^2$$

so n_g (1550 nm) = 4.216 in agreement with the group index solved directly by MODE at 1550 nm which is 4.215.

The mode solver results were exported as a Lumerical Data File (.ldf), which holds the effective index and group index of the waveguide as a function of wavelength. This file is loaded into the INTERCONNECT waveguide elements, so the circuit model uses the same dispersion as the mode solution.

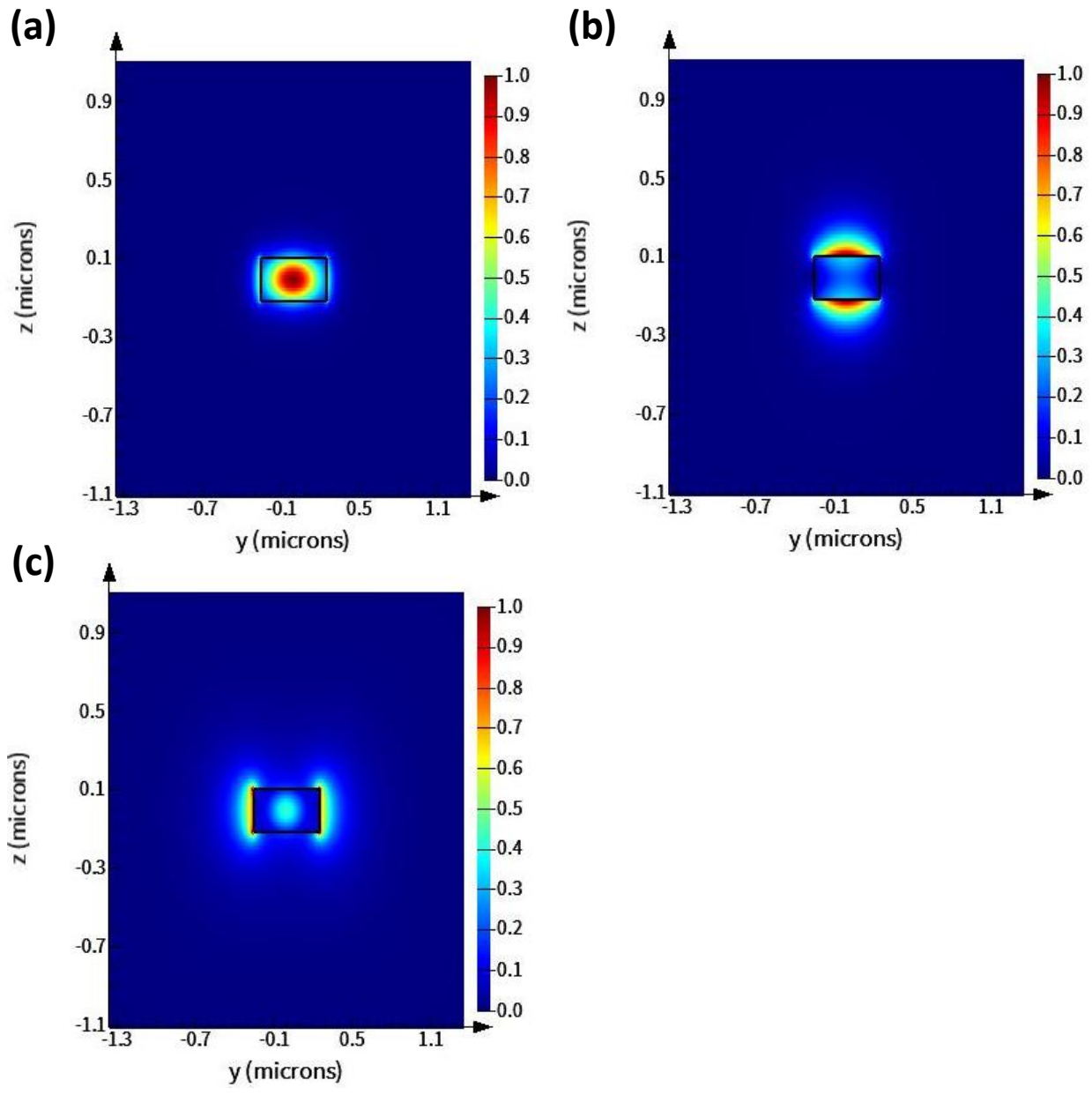


Figure 2. Electric-field intensity of the three modes found by the mode solver at 1550 nm for the 500 × 220 nm silicon waveguide. (a) Mode 1: TE fundamental mode, n_{eff} = 2.445, TE fraction 98%; the field is confined in the silicon core. (b) Mode 2: TM-like mode, n_{eff} = 1.772, TE fraction 4%. (c) Mode 3: n_{eff} = 1.493, close to the silicon dioxide cladding index; the field is not confined by the core.

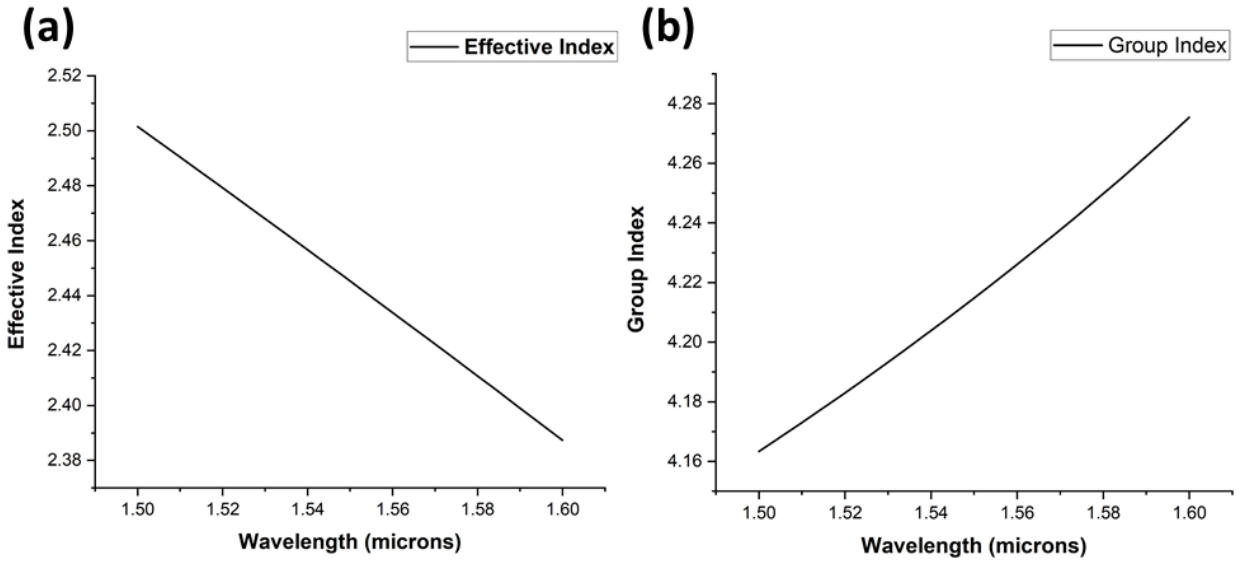


Figure 3. (a) Effective index and (b) group index of the fundamental TE mode as a function of wavelength from 1500 to 1600 nm. The group index at 1550 nm is 4.216.

The layout (Figure 4) contains the loopback reference and the four MZIs. The circuit of each MZI was extracted from the KLayout layout using the SiEPIC circuit simulation tool and simulated in Lumerical INTERCONNECT (Figure 5, shown for MZI4). The grating couplers, the Y-branch splitter and the broadband directional coupler used the compact models of the PDK. In the two MZI arms, the standard PDK waveguide, whose internal model cannot be modified, was replaced by an INTERCONNECT waveguide element that reads the effective-index and group-index data exported from the MODE simulation as mentioned earlier, so that the interferometer uses the same dispersion as the waveguide model. Each arm was given its length from the layout (Table 1) and a propagation loss of 300 dB/m (3 dB/cm). The short routing waveguides between the output coupler and the grating couplers keep the PDK model; they lie outside the interferometer and do not affect ΔL or the FSR. The transmission of the two outputs was computed for TE polarisation from 1499 to 1601 nm in 5000 points, a step of about 20 pm.

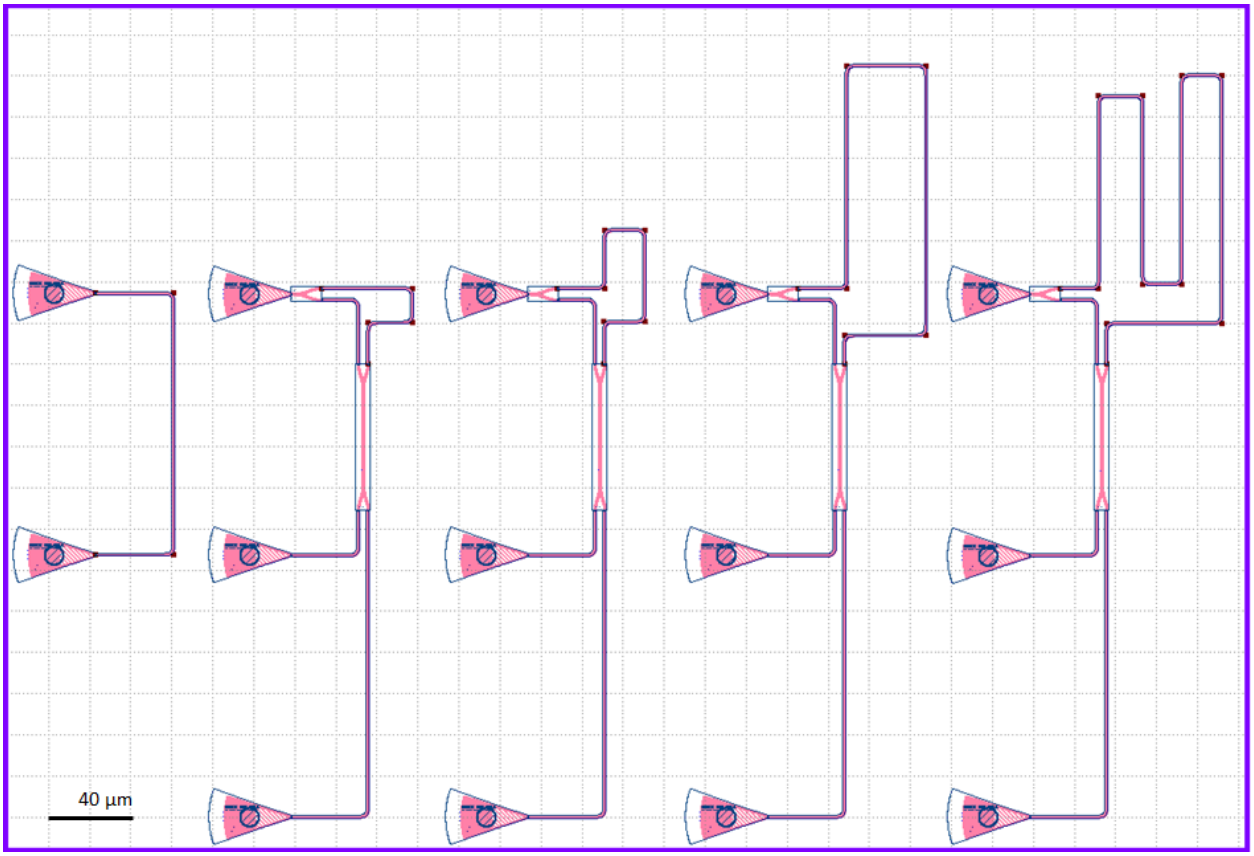


Figure 4. Layout of the submitted design (KLayout), showing from left to right the loopback reference and the four unbalanced Mach–Zehnder interferometers MZI1 to MZI4, with path differences ΔL of 50.000, 100.000, 300.000 and 500.000 µm. Light enters each MZI through the top grating coupler and is split by a Y-branch into two arms of different lengths. The arms are recombined in a broadband directional coupler, whose two outputs go to the two lower grating couplers. The loopback is a pair of grating couplers joined by a waveguide and is used to measure the loss of the grating couplers and routing alone. Scale bar: 40 µm.

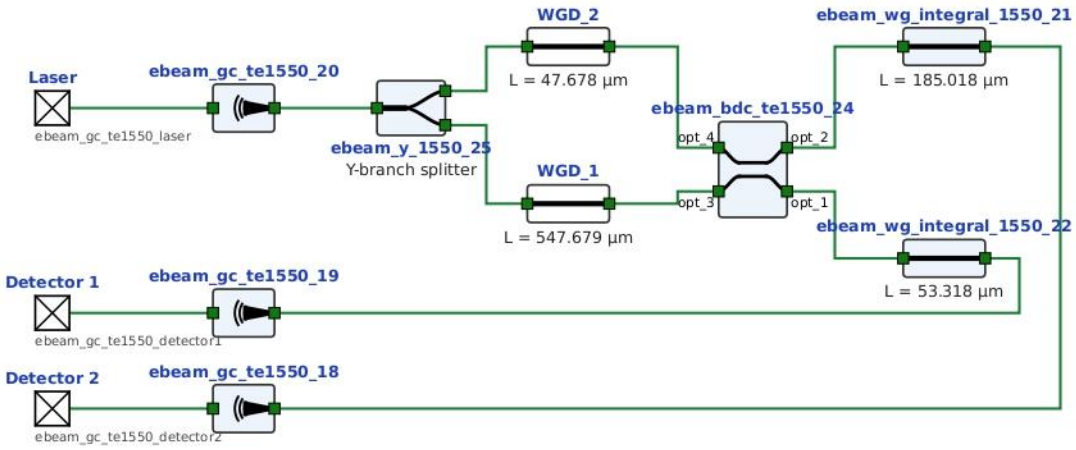


Figure 5. INTERCONNECT circuit model of MZI4. Light from the laser enters through a grating coupler and is split by a Y-branch into two arms of length 47.678 µm and 547.679 µm, giving a path difference ΔL of 500.001 µm. The arms are waveguide elements that use the group index from the mode solution. They are recombined in a

broadband directional coupler, and its two outputs go through short routing waveguides to the grating couplers at detectors 1 and 2. MZI1 to MZI3 use the same circuit with different arm lengths (Table 1).

Results

Figure 6(a–d) shows the simulated transmission of the four MZIs over 1499–1601 nm. Each spectrum contains two traces, one for each output of the broadband directional coupler. The number of interference fringes in the band increases with the path difference, from about 9 nulls per output for MZI1 to about 90 for MZI4, which is the width of the band divided by the FSR. The two outputs are complementary: a minimum in one output coincides with a maximum in the other. The slow variation of the overall transmission, with a maximum of about −5 to −6 dB near the centre of the band and a decrease towards the edges, is mainly the response of the grating couplers and is the same for all four devices.

The free spectral range (FSR) was extracted from the INTERCONNECT transmission spectra using the positions of the transmission minima (nulls). Near a null the transmitted power is approximately a parabola in wavelength, so a quadratic polynomial was fitted, on a linear power scale, to the seven data points (about ±60 pm) around each minimum, and the vertex of the parabola was taken as the null wavelength. The FSR was defined as the spacing between adjacent nulls of the same output port, assigned to their midpoint wavelength. A quadratic fit of FSR against wavelength was then evaluated at 1550 nm, the design wavelength. Nulls were used instead of peaks because the grating-coupler envelope shifts the apparent peak positions, whereas the nulls are narrow and less affected by it.

The simulated FSR agrees with the formula to within 2.2% for all four devices (Table 2), and the agreement improves as the path difference grows: the simulated value is 2.14% below the formula for MZI1 (ΔL = 50.000 µm) and 0.20% below for MZI4 (ΔL = 500.000 µm). Both columns use the same MODE waveguide data, so the difference does not come from the waveguide model itself. It also cannot be an error in n_g, since an error in n_g would give the same percentage difference for all four devices, whereas the difference falls from 2.1% to 0.2% as ΔL grows from 50 to 500 µm. The difference falls roughly as 1/ΔL, as expected for a constant extra path length, and corresponds to roughly an extra path of 1.0–1.1 µm in every device. Its origin was not investigated further.

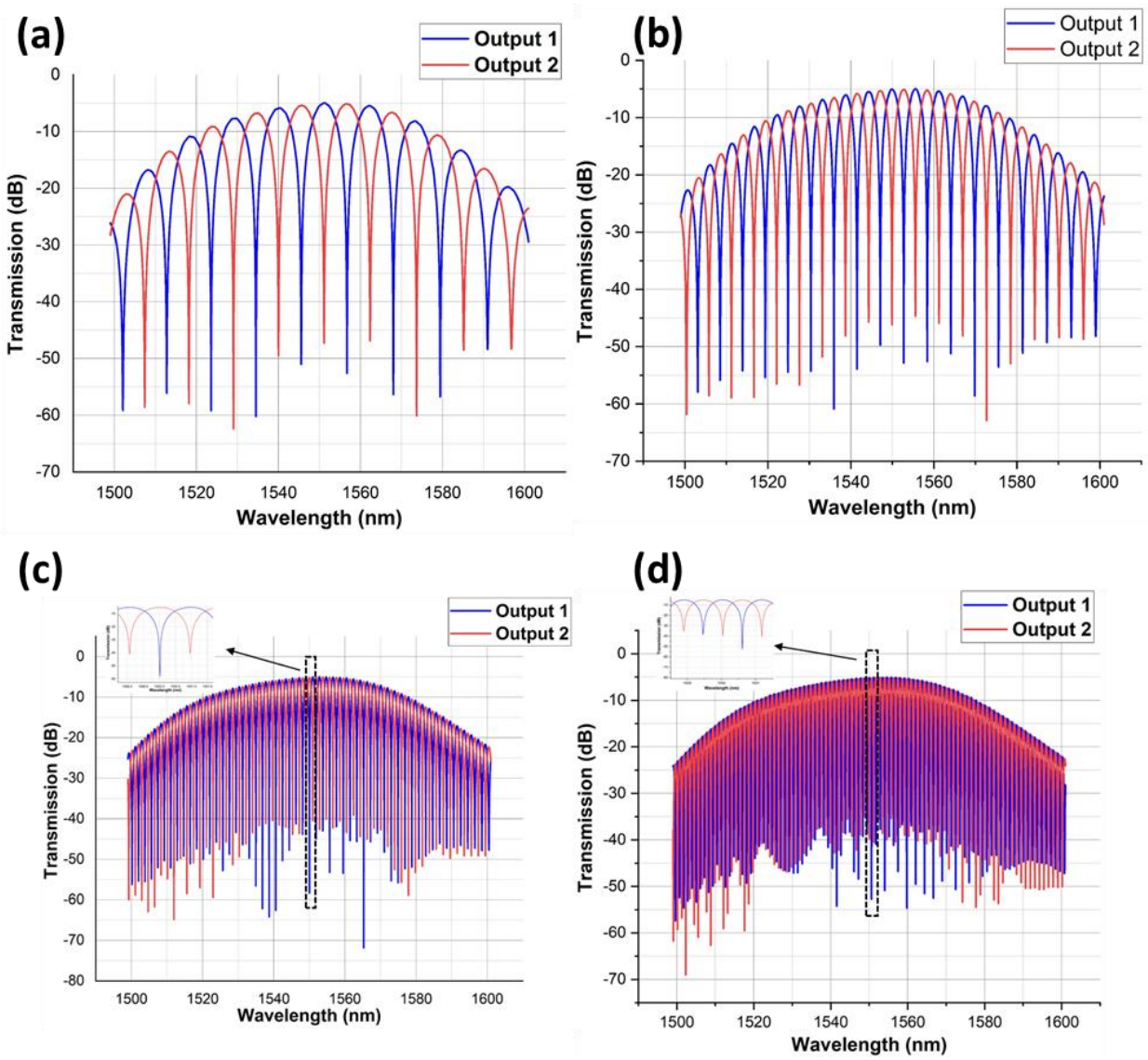


Figure 6. Simulated transmission spectra of the two outputs of the four unbalanced MZIs, calculated in INTERCONNECT for TE polarization from 1499 to 1601 nm in 20 pm steps: (a) MZI1, ΔL = 50.000 µm; (b) MZI2, ΔL = 100.000 µm; (c) MZI3, ΔL = 300.000 µm; (d) MZI4, ΔL = 500.000 µm. The two outputs (blue and red) are complementary: the peaks of one coincide with the nulls of the other. The fringe spacing decreases as ΔL increases, with simulated FSRs at 1550 nm of 11.15, 5.64, 1.89 and 1.14 nm respectively (Table 2). The smooth envelope is set by the grating couplers. The insets in (c) and (d) magnify the region around 1550 nm marked by the dashed box. The depth of the nulls is limited by the 20 pm wavelength step and is not a physical extinction ratio.

Table 2. Free spectral range at 1550 nm from the formula, the INTERCONNECT simulation and measurement.

Device	ΔL (µm)	Formula FSR (nm)	Simulated FSR (nm)	Formula FSR - Simulated FSR (nm)	Difference (%)	Measured FSR from experiment (nm)
MZI1	50.000	11.398	11.153	0.245	2.14	*to be measured*
MZI2	100.000	5.699	5.638	0.061	1.07	*to be measured*
MZI3	300.000	1.8997	1.8932	0.0065	0.34	*to be measured*
MZI4	500.000	1.1398	1.1375	0.0023	0.20	*to be measured*

Experiment data

To be completed after measurement: the output spectra of the loopback device and MZI1 to MZI4 and a description of the measurement setup.

Analysis

To be completed: FSR extraction, n_g from each device and from the 1/FSR fit, comparison with the simulated n_g, and the effect of fabrication variation.

Conclusion

To be completed.

References

[1] S. A. Al Graiti and S. A. Jasim, "Characterizing polarization-based designs of unbalanced Mach-Zehnder interferometers," *Kufa Journal of Engineering*, vol. 17, no. 3, pp. 652-664, 2026, doi:10.30572/2018/KJE/170337